

ML Application to IoT

Dr. Bharti Trivedi & Ms.Shweta Rajput

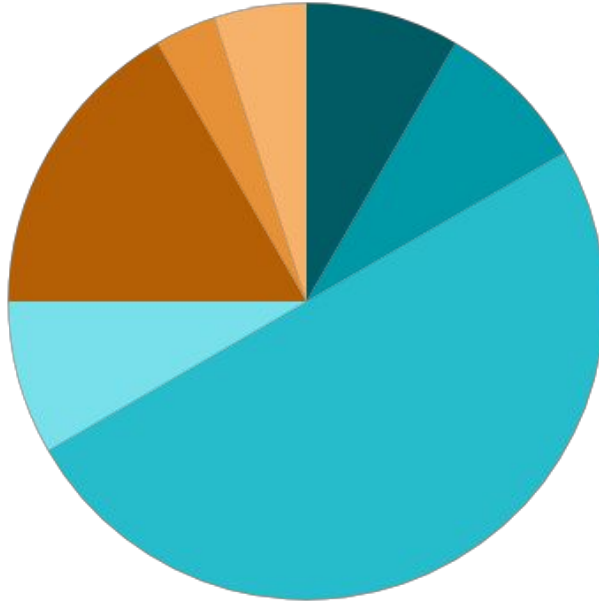
Department of Computer Science & Engineering



Outline

1. Real time tracking and optimization of logistics and public transportation Systems - <https://www.geios.com/read/2VSPI8>
2. <https://github.com/salildz/vehicle-tracking>
- 3.
4. Remote inspection and assessment of damage and accidents
5. Chronic disease management using remote expert consultation

Lecture Structure



05 Minutes

Outline & Objectives



05 Minutes

Recap of Previous Lecture



30 Minutes

Delivery of Actual Content as per Lesson Plan



05 Minutes

Industry Relevance



10 Minutes

Interactive Activities



02 Minutes

Importance of Discipline



03 Minutes

Recording of Attendance

Reference: Office Order No. Provost/092023135, Date: 18th September 2023



Objectives

1. Understand how ML adds intelligence to IoT systems.
2. Differentiate static data vs. real-time IoT data.
3. Explain IoT-ML workflow: Sensors → Cloud → Model → Action.
4. Apply ML in logistics, inspection, and healthcare IoT use cases.
5. Identify key challenges: noise, latency, scalability, privacy.
6. Design simple IoT-ML prototypes for prediction and automation.

Introduction - IoT

Definition:

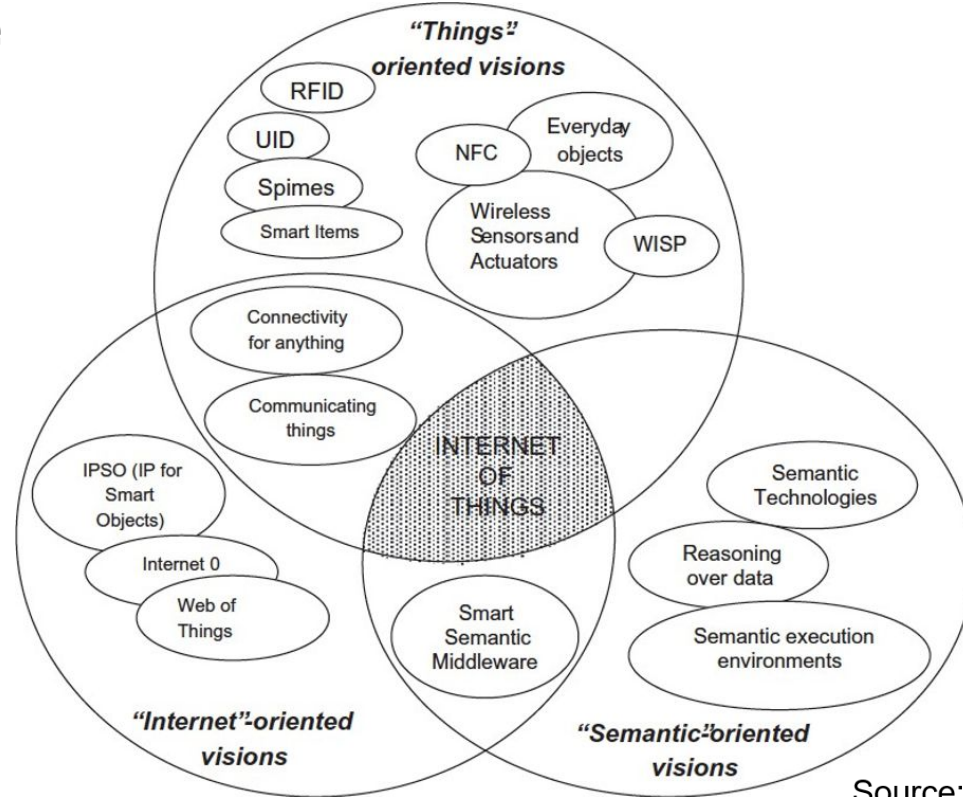
A network of interconnected devices that **collect, transmit, and act on data** from the physical world using sensors and communication networks.

Key Components:

- **Sensors & Actuators:** Measure and act on environment
- **Connectivity:** Wi-Fi, Bluetooth, Zigbee, LoRa, 4G/5G
- **Data Processing:** Edge devices or cloud computing
- **User Interface:** Dashboards, mobile apps, alerts



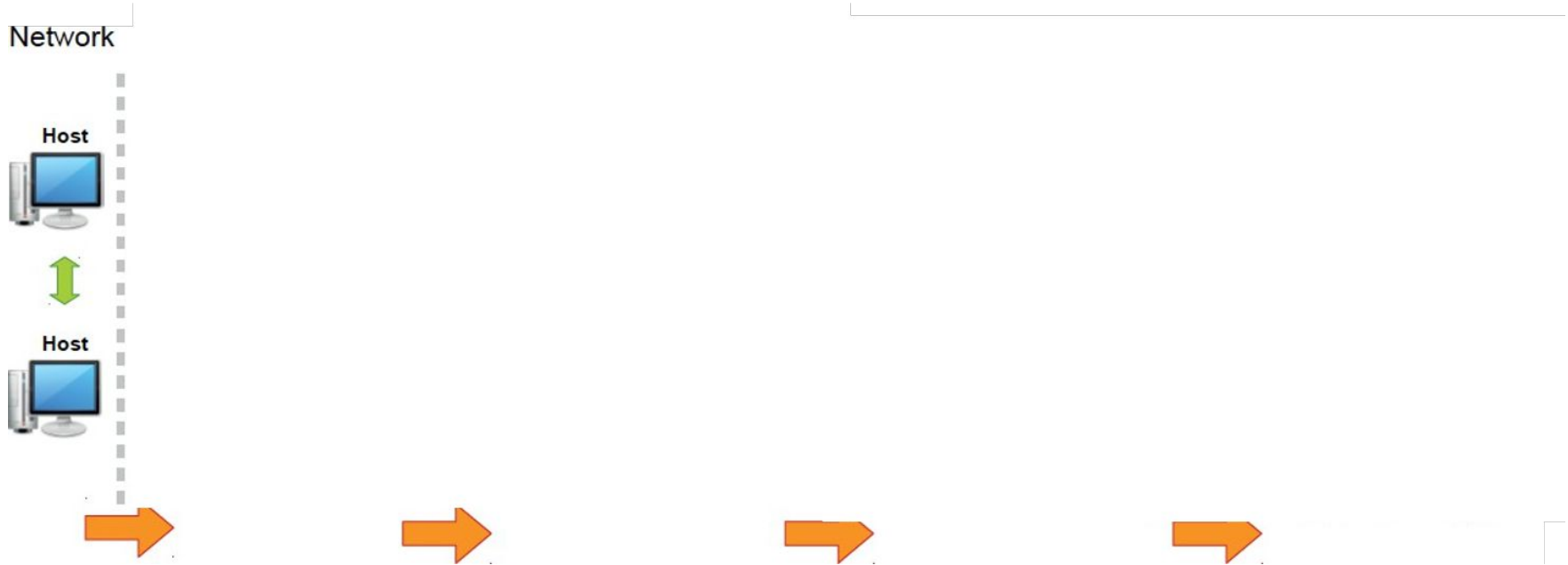
Connection of Multiple



Source: Atzori et al. 2010



IoT Evolution

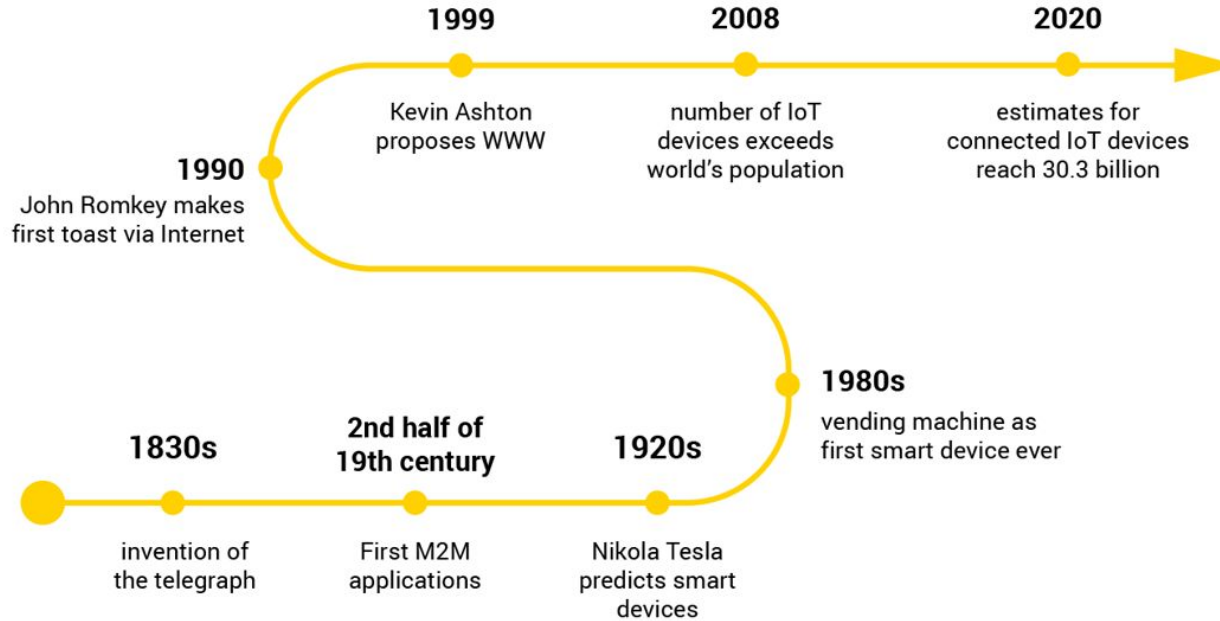


Starts with only network and evolves into everything that can be connected with *a network*

Source: Perera et al. 2014



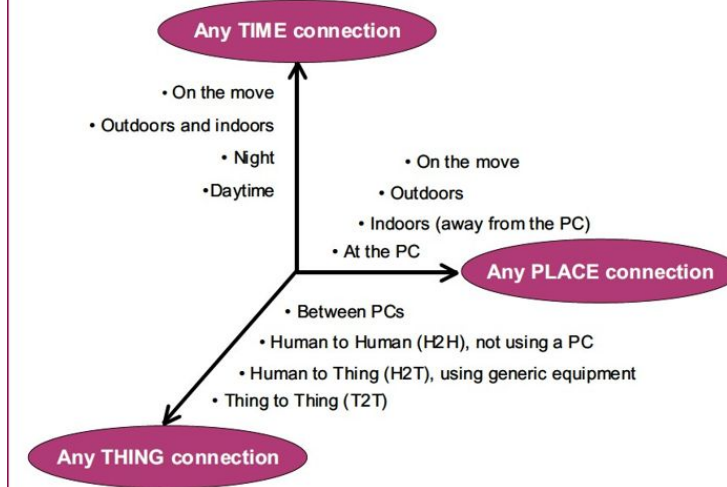
History of IoT



Any-X Point of View



Figure 1 – A new dimension



Source: ITU adapted from Nomura Research Institute

The Internet of Things allows people and things to be connected Anytime, Anyplace, with Anything and Anyone, ideally using Any path/ network and Any service.

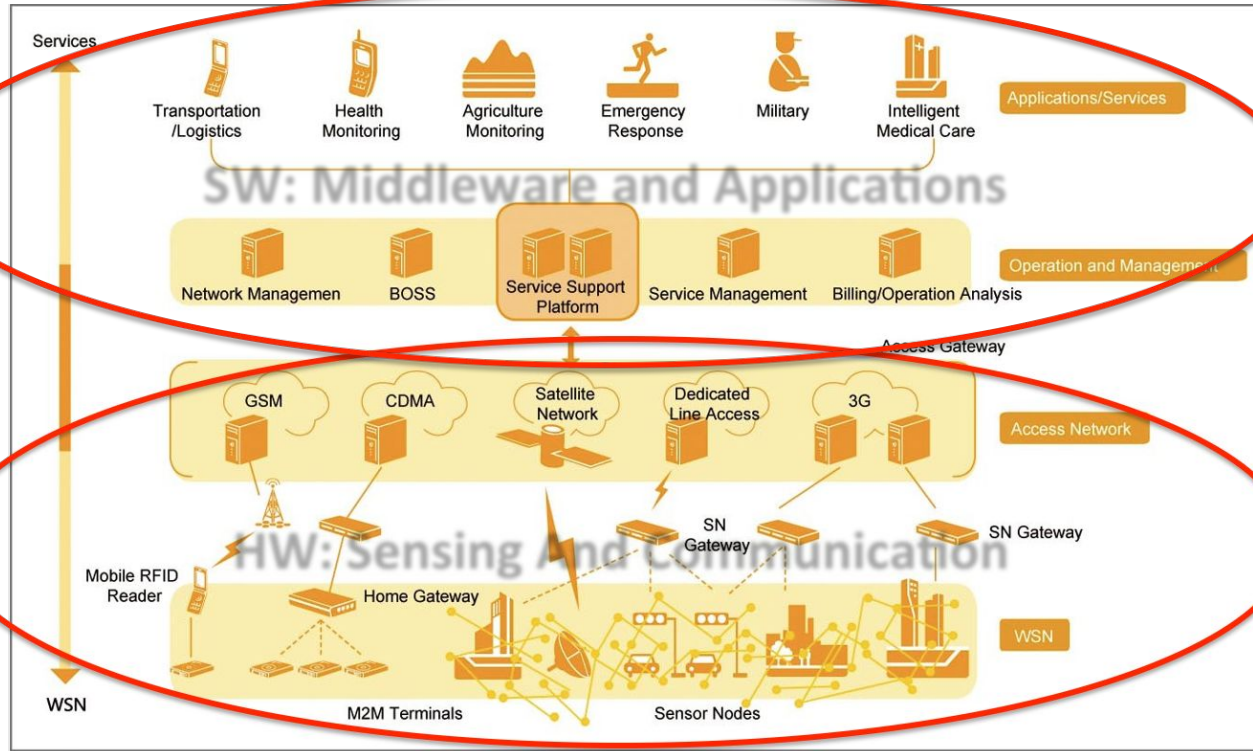
Source: Perera et al. 2014



Characteristics of IoT

1. Intelligence – Knowledge extraction from the generated data
2. Architecture – A hybrid architecture supporting many others
3. Complex system – A diverse set of dynamically changing objects
4. Size considerations – Scalability
5. Time considerations – Billions of parallel and simultaneous events
6. Space considerations – Localization
7. Everything-as-a-service – Consuming resources as a service

IoT Layered Architecture



Architecture of Internet of Things (IoT)

Stage 1: Sensing Layer (Data Collection)

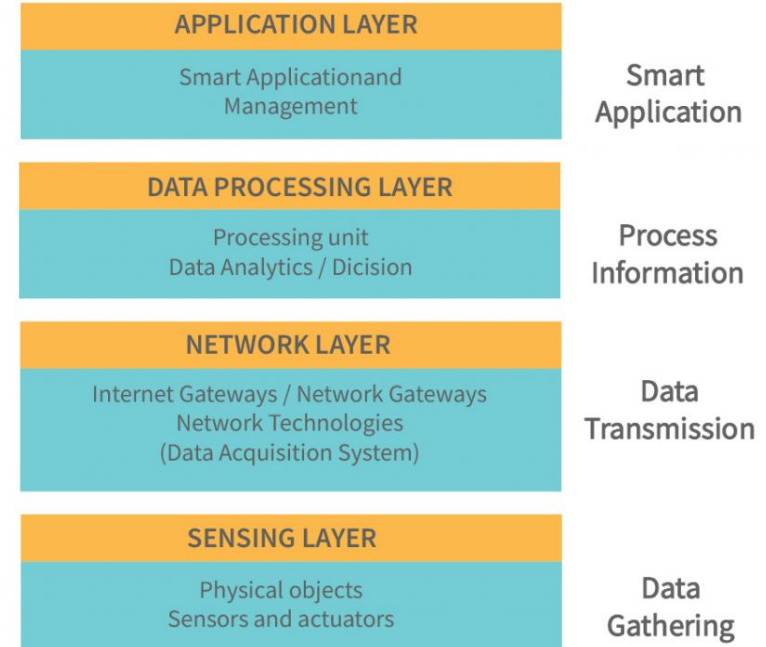
- Devices/sensors collect raw data from the environment.
- Examples:
 - GPS → location
 - Temperature sensor → environment
 - Accelerometer → movement
 - Camera → image data
- Hardware: Arduino, Raspberry Pi, ESP32, smart sensors.

Keywords: Analog → Digital conversion, continuous data streams.

Data Flow
Control Flow



4- STAGE LOT ARCHITECTURE



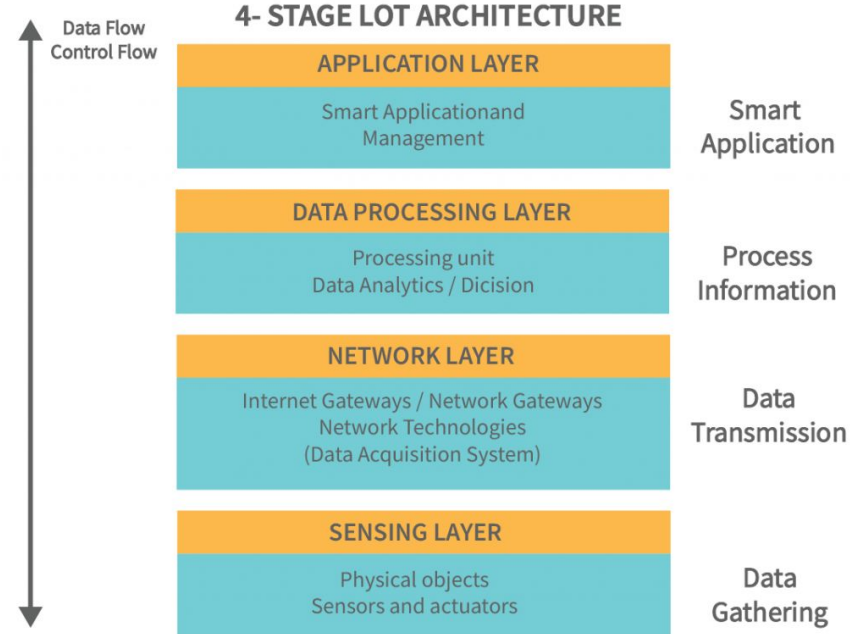
<https://www.interviewbit.com/blog/iot-architecture/>

Architecture of Internet of Things (IoT)

Stage 2: Network Layer (Data Transmission)

- Transfers sensor data to the cloud or local server.
- Uses wireless protocols:
 - Wi-Fi, Bluetooth, Zigbee, LoRaWAN, MQTT, 4G/5G.
- Gateways aggregate data from multiple IoT nodes.

Concepts to teach: latency, bandwidth, real-time transmission.



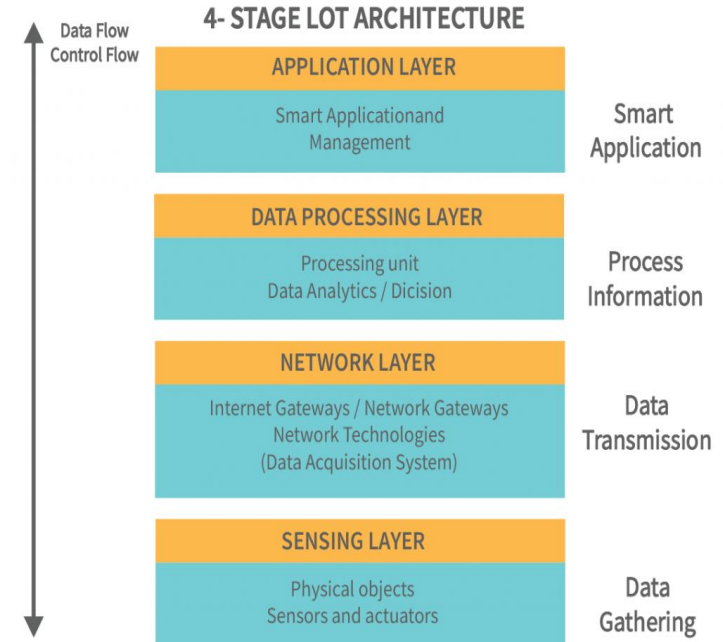
<https://www.interviewbit.com/blog/iot-architecture/>

Architecture of Internet of Things (IoT)

Stage 3: Data Processing / Cloud Layer

- Raw data is stored and cleaned.
- ML model is trained (offline) or used (online inference).
- Cloud platforms like:
 - **AWS IoT Core, Google Cloud IoT, Azure IoT Hub.**
- Real-time processing tools:
 - Kafka, Spark Streaming, AWS Lambda.

Keywords: preprocessing, feature extraction, batch vs. stream processing.

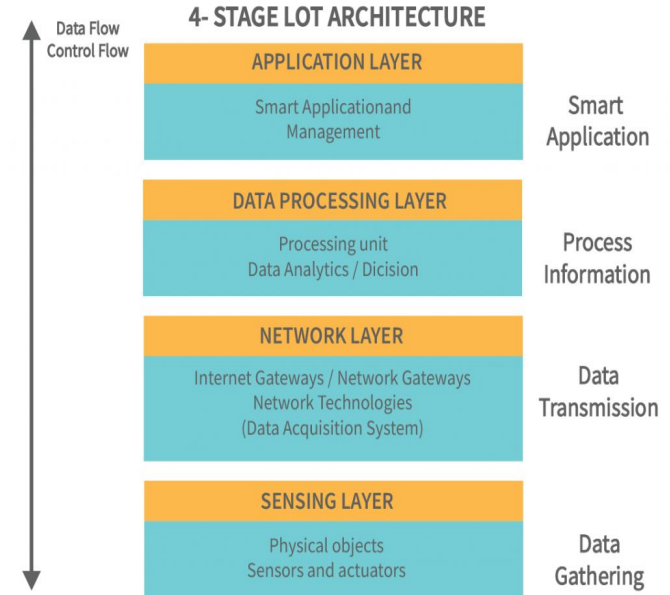


<https://www.interviewbit.com/blog/iot-architecture/>

Architecture of Internet of Things (IoT)

Machine Learning / Intelligence Layer

- ML models analyze incoming data to **detect, predict, or classify**.
- Common ML tasks:
 - Regression:** Predict numeric values (e.g., temperature next hour)
 - Classification:** Detect event (normal/abnormal)
 - Clustering:** Group similar patterns
 - Anomaly detection:** Detect failures or outliers
- Examples:
 - Predict when a motor will fail based on vibration data.
 - Detect traffic congestion from GPS patterns.



<https://www.interviewbit.com/blog/iot-architecture/>



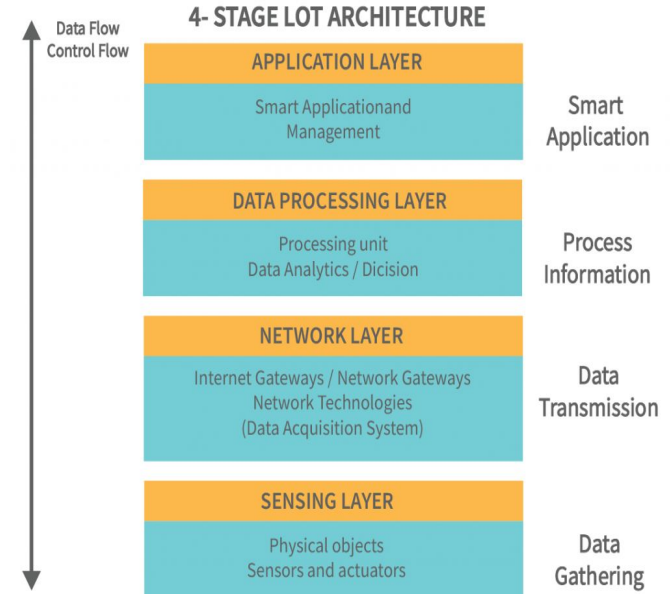
Architecture of Internet of Things (IoT)

Stage 4: Action / Application Layer

- ML model outputs trigger actions:
 - Send alerts, adjust device settings, or control actuators.
- Actions can be:
 - **Automated:** e.g., turn off a machine before overheating.
 - **Human-assisted:** e.g., notify operator for inspection.

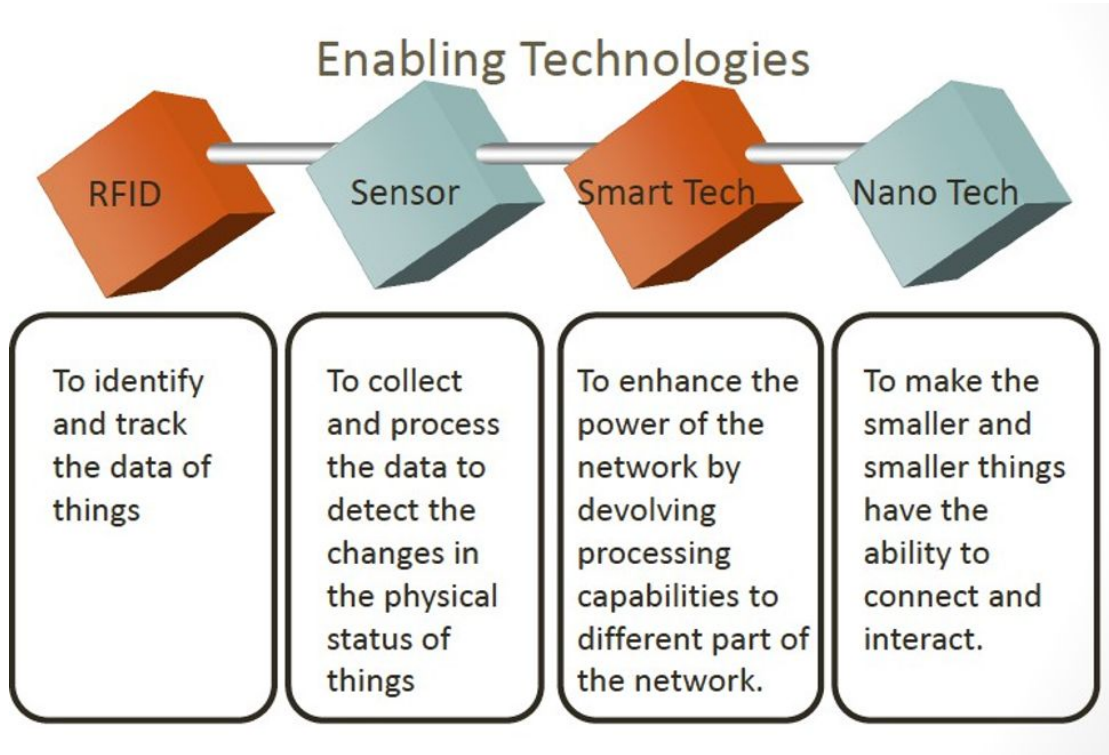
Output can go to:

- Mobile App dashboard (IoT visualization)
- Cloud-based monitoring panel
- Actuator (fan, alarm, relay, etc.)



<https://www.interviewbit.com/blog/iot-architecture/>

Networking and Communication

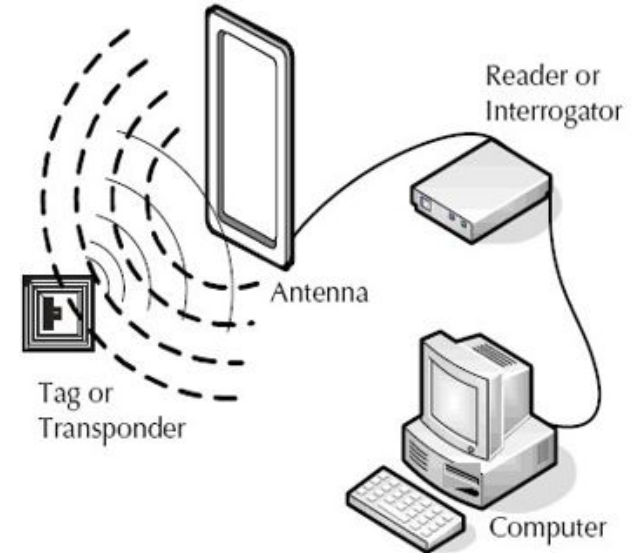


- RFID to smallest enabling technologies, such as chips, etc.
- Mobile platforms, such as sensors, phones, etc.

Source: Qian Zhang. Lecture notes. 2013

RFIDs

- The reduction in terms of size, weight, energy consumption, and cost of the radio takes us to a new era
 - This allows us to integrate radios in almost all objects and thus, to add the world “anything” to the above vision which leads to the IoT concept
- Composed of one or more readers and tags
- RFID tag is a small microchip attached to an antenna
- Can be seen as one of the main, smallest components of IoT, that collects data





Wireless Technologies

- Telecommunication systems
 - Initial/primary service: mobile voice telephony
 - Large coverage per access point (100s of meters – 10s of kilometers)
 - Low/moderate data rate (10s of kbit/s – 10s of Mbits/s)
 - Examples: GSM, UMTS, LTE
- WLAN
 - Initial service: Wireless Ethernet extension
 - Moderate coverage per access point (10s – 100s meters)
 - Moderate/high data rate (Mbits/s – 100s)
 - Examples: IEEE 802.11(a-g), Wimax

Source: Geert Heijenk. Mobile and Wireless Networking. Lecture Notes. 2014



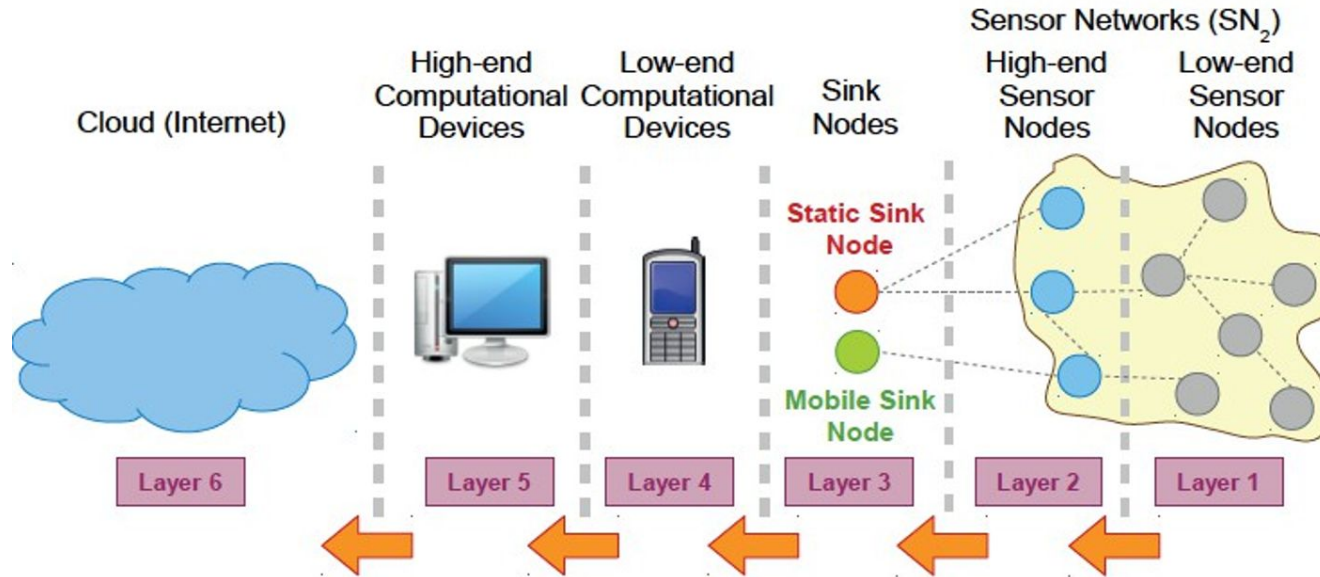
Wireless Technologies

- Short range:
 - Direct connection between devices – sensor networks
 - Typical low power usage
 - Examples: Bluetooth, Zigbee, Z-wave (house products)
- Other examples:
 - Satellite systems
 - Global coverage
 - Applications: audio/TV broadcast, positioning, personal communications
 - Broadcast systems
 - Satellite/terrestrial
 - Support for high speed mobiles
 - Fixed wireless access
 - Several technologies including DECT, WLAN, IEEE802.16, etc.

Source: Geert Heijenk. *Mobile and Wireless Networking. Lecture Notes. 2014*

Sensor Networks (SNs)

Consist of a certain number (which can be very high) of sensing nodes generally wireless) communicating in a wireless multi-hop fashion



Source: Perera et al. 2014



Data Flow Example

Example: Smart Vehicle Monitoring System

1. **Sensor:** GPS + Accelerometer track speed and motion
2. **Network:** Sends data to cloud using MQTT
3. **Processing:** Cloud stores data in database
4. **ML Model:** Predicts fuel consumption & detects rash driving
5. **Action:** Sends alert to driver or updates dashboard

<https://www.interviewbit.com/blog/iot-architecture/>

Integration Approaches

Type	Description	Example
Cloud ML	Train + run model in cloud	AWS SageMaker, Google Cloud AI
Edge ML	Run lightweight ML model on IoT device	TinyML on Arduino, TensorFlow Lite on Pi
Hybrid	Train in cloud, deploy inference model on edge	Predict locally, sync periodically

<https://www.interviewbit.com/blog/iot-architecture/>

Common IoT Boards & Their Utilization

Board	Type / Specs	Typical Use Case
Arduino Uno / Nano	Microcontroller (8-bit)	Basic sensor interfacing, LED control, small IoT prototypes
ESP8266 / ESP32	Microcontroller with Wi-Fi (ESP32 also has Bluetooth)	IoT projects needing internet connectivity (e.g., home automation, data to cloud)
Raspberry Pi	Mini computer (Linux-based)	Advanced IoT + ML applications, camera projects, edge computing
NodeMCU	Based on ESP8266	Low-cost IoT Wi-Fi board for beginners
Jetson Nano / Raspberry Pi 4 + AI	Edge AI hardware	Real-time object detection, ML inference on device
STM32 / BeagleBone	Industrial-grade microcontrollers	Robotics, industrial automation, precision IoT control

Common IoT Boards & Their Utilization



Arduino Uno



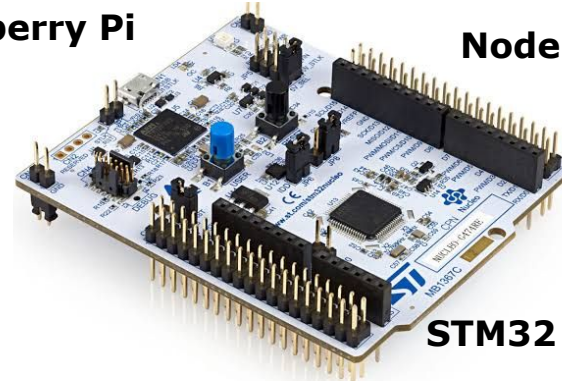
Raspberry Pi



NodeMCU



Jetson Nano



STM32

Difference Between IoT Data and Static ML Data

Feature	Static ML Data	IoT Data (Real-Time / Sensor / Streaming)
Nature of Data	Fixed, pre-collected dataset	Continuously generated from sensors/devices
Data Flow	Batch — data is loaded once	Stream — data arrives every second or millisecond
Example Sources	CSV file, Kaggle dataset, image folder	Temperature sensors, GPS trackers, smart meters, wearables
Volume & Velocity	Moderate, easy to handle	High volume, high velocity (“Big Data”)
Structure	Often clean, structured tables	Often unstructured or semi-structured (JSON, MQTT messages, logs)
Update Frequency	Collected once → static	Constantly updating, time-stamped data

Difference Between IoT Data and Static ML Data

Feature	Static ML Data	IoT Data (Real-Time / Sensor / Streaming)
Noise Level	Usually cleaned before training	Very noisy — missing values, sensor drift, false readings
Storage	Stored locally for training	Stored in time-series databases (InfluxDB, AWS Timestream, MongoDB)
Model Interaction	Offline — train → test → done	Online / real-time inference and retraining needed
Computation Location	Centralized (on PC/Cloud)	Distributed (on edge devices, gateways, cloud)
Example Scenario	Predict house prices using static dataset	Predict traffic congestion from live GPS sensor data



Application

<https://github.com/PacktPublishing/Hands-On-Artificial-Intelligence-for-IoT/blob/master/Chapter12/video.avi>

<https://www.interviewbit.com/blog/iot-architecture/>

 **Thank You..**